

Lab Assignment 4: Creation and Insertion in University Database.

Database Description: A university database system maintains information about instructor, students, department, course, teaches, classroom, section, and takes. In this assignment, students will create tables using database constraints and insert sample records.

Objective: To understand SQL commands for table creation (CREATE) and data insertion (INSERT) using database constraints.

1. Create the following tables with their respective columns, data types and size:

Table Name	Columns	CONSTRAINT DESCRIPTION
DEPARTMENT	DEPT_NAME	-PRIMARY KEY
	BUILDING	-NOT NULL
	BUDGET	-MUST BE > 0

Table Name	Columns	CONSTRAINT DESCRIPTION
INSTRUCTOR	ID	-PRIMARY KEY -LENGTH MUST BE 5
	NAME	-NOT NULL
	DEPT_NAME	-FOREIGN KEY, REFERS TABLE DEPARTMENT
	SALARY	-MUST BE > 20000

Table Name	Columns	CONSTRAINT DESCRIPTION
STUDENT	ID	-PRIMARY KEY -LENGTH MUST BE 5
	NAME	-NOT NULL
	DEPT_NAME	-FOREIGN KEY, REFERS TABLE DEPARTMENT
	TOT_CRED	-MUST BE > =0

Table Name	Columns	CONSTRAINT DESCRIPTION
ADVISOR	I_ID	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE INSTRUCTOR
	S_ID	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE STUDENT

Table Name	Columns	CONSTRAINT DESCRIPTION
COURSE	COURSE_ID	-PRIMARY KEY
	TITLE	-NOT NULL
	DEPT_NAME	-FOREIGN KEY, REFERS TABLE DEPARTMENT
	CREDIT	-MUST BE > 0

Table Name	Columns	CONSTRAINT DESCRIPTION
CLASSROOM	ROOM_NO	-PRIMARY KEY
	BUILDING	-PRIMARY KEY
	CAPACITY	-MUST BE > =10

Table Name	Columns	CONSTRAINT DESCRIPTION
SECTION	COURSE_ID	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE COURSE
	SEC_ID	-PRIMARY KEY -NOT NULL
	SEMESTER	-PRIMARY KEY -ANY OF SPRING, SUMMER, FALL, WINTER
	YEAR	-PRIMARY KEY -MUST BE >=2000 AND <= 2026
	ROOM_NO	-FOREIGN KEY, REFERS TABLE CLASSROOM
	BUILDING	-FOREIGN KEY, REFERS TABLE CLASSROOM

Table Name	Columns	CONSTRAINT DESCRIPTION
TEACHES	ID	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE INSTRUCTOR
	COURSE_ID	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE SECTION
	SEC_ID	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE SECTION
	SEMESTER	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE SECTION
	YEAR	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE SECTION

Table Name	Columns	CONSTRAINT DESCRIPTION
TAKES	S_ID	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE STUDENT
	COURSE_ID	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE SECTION
	SEC_ID	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE SECTION
	SEMESTER	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE SECTION
	YEAR	-PRIMARY KEY -FOREIGN KEY, REFERS TABLE SECTION
	GRADE	

```

mysql> CREATE DATABASE 2341004117_ASSIGNMENT_4;
Query OK, 1 row affected (0.02 sec)

mysql> USE 2341004117_ASSIGNMENT_4;
Database changed
mysql> CREATE TABLE DEPARTMENT (
->     DEPT_NAME VARCHAR(50) PRIMARY KEY,
->     BUILDING VARCHAR(50) NOT NULL,
->     BUDGET DECIMAL(10,2) CHECK (BUDGET > 0)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE INSTRUCTOR (
->     ID CHAR(5) PRIMARY KEY,
->     NAME VARCHAR(50) NOT NULL,
->     DEPT_NAME VARCHAR(50),
->     SALARY DECIMAL(10,2) CHECK (SALARY > 20000),
->     FOREIGN KEY (DEPT_NAME) REFERENCES DEPARTMENT(DEPT_NAME)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE STUDENT (
->     ID CHAR(5) PRIMARY KEY,
->     NAME VARCHAR(50) NOT NULL,
->     DEPT_NAME VARCHAR(50),
->     TOT_CRED INT CHECK (TOT_CRED >= 0),
->     FOREIGN KEY (DEPT_NAME) REFERENCES DEPARTMENT(DEPT_NAME)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE ADVISOR (
->     I_ID CHAR(5),
->     S_ID CHAR(5),
->     PRIMARY KEY (I_ID, S_ID),
->     FOREIGN KEY (I_ID) REFERENCES INSTRUCTOR(ID),
->     FOREIGN KEY (S_ID) REFERENCES STUDENT(ID)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE COURSE (
->     COURSE_ID VARCHAR(10) PRIMARY KEY,
->     TITLE VARCHAR(100) NOT NULL,
->     DEPT_NAME VARCHAR(50),
->     CREDIT INT CHECK (CREDIT > 0),
->     FOREIGN KEY (DEPT_NAME) REFERENCES DEPARTMENT(DEPT_NAME)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE CLASSROOM (
->     ROOM_NO INT,
->     BUILDING VARCHAR(50),
->     CAPACITY INT CHECK (CAPACITY >= 10),
->     PRIMARY KEY (ROOM_NO, BUILDING)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE SECTION (
->     COURSE_ID VARCHAR(10),
->     SEC_ID INT,
->     SEMESTER VARCHAR(10),
->     YEAR INT,
->     ROOM_NO INT,
->     BUILDING VARCHAR(50),
->     PRIMARY KEY (COURSE_ID, SEC_ID, SEMESTER, YEAR),
->     FOREIGN KEY (COURSE_ID) REFERENCES COURSE(COURSE_ID),
->     FOREIGN KEY (ROOM_NO, BUILDING) REFERENCES CLASSROOM(ROOM_NO, BUILDING),
->     CHECK (SEMESTER IN ('Spring','Summer','Fall','Winter')),
->     CHECK (YEAR >= 2000 AND YEAR <= 2026)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE TEACHES (
->     ID CHAR(5),
->     COURSE_ID VARCHAR(10),
->     SEC_ID INT,
->     SEMESTER VARCHAR(10),
->     YEAR INT,
->     PRIMARY KEY (ID, COURSE_ID, SEC_ID, SEMESTER, YEAR),
->     FOREIGN KEY (ID) REFERENCES INSTRUCTOR(ID),
->     FOREIGN KEY (COURSE_ID, SEC_ID, SEMESTER, YEAR)
->         REFERENCES SECTION(COURSE_ID, SEC_ID, SEMESTER, YEAR)
-> );
Query OK, 0 rows affected (0.03 sec)

```

```
mysql> CREATE TABLE TAKES (
->   S_ID CHAR(5),
->   COURSE_ID VARCHAR(10),
->   SEC_ID INT,
->   SEMESTER VARCHAR(10),
->   YEAR INT,
->   GRADE VARCHAR(2),
->   PRIMARY KEY (S_ID, COURSE_ID, SEC_ID, SEMESTER, YEAR),
->   FOREIGN KEY (S_ID) REFERENCES STUDENT(ID),
->   FOREIGN KEY (COURSE_ID, SEC_ID, SEMESTER, YEAR)
->     REFERENCES SECTION(COURSE_ID, SEC_ID, SEMESTER, YEAR)
-> );
Query OK, 0 rows affected (0.03 sec)
```

2. Insert data into Department table with the given data.

<i>dept_name</i>	<i>building</i>	<i>budget</i>
Biology	Watson	90000
Comp. Sci.	Taylor	100000
Elec. Eng.	Taylor	85000
Finance	Painter	120000
History	Painter	50000
Music	Packard	80000
Physics	Watson	70000

```
mysql> INSERT INTO DEPARTMENT VALUES
-> ('Biology','Watson',90000),
-> ('Comp. Sci.','Taylor',100000),
-> ('Elec. Eng.','Taylor',85000),
-> ('Finance','Painter',120000),
-> ('History','Painter',50000),
-> ('Music','Packard',80000),
-> ('Physics','Watson',70000);
Query OK, 7 rows affected (0.02 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

3. Insert data into Instructor table with the given data.

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

```
mysql> INSERT INTO INSTRUCTOR VALUES
-> ('10101','Srinivasan','Comp. Sci.',65000),
-> ('12121','Wu','Finance',90000),
-> ('15151','Mozart','Music',40000),
-> ('22222','Einstein','Physics',95000),
-> ('32343','El Said','History',60000),
-> ('33456','Gold','Physics',87000),
-> ('45565','Katz','Comp. Sci.',75000),
-> ('58583','Califieri','History',62000),
-> ('76543','Singh','Finance',80000),
-> ('76766','Crick','Biology',72000),
-> ('83821','Brandt','Comp. Sci.',92000),
-> ('98345','Kim','Elec. Eng.',80000);
Query OK, 12 rows affected (0.01 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

4. Insert data into Student table with the given data.

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>tot_cred</i>
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

```
mysql> INSERT INTO STUDENT VALUES
-> ('00128','Zhang','Comp. Sci.',102),
-> ('12345','Shankar','Comp. Sci.',32),
-> ('19991','Brandt','History',80),
-> ('23121','Chavez','Finance',110),
-> ('44553','Peltier','Physics',56),
-> ('45678','Levy','Physics',46),
-> ('54321','Williams','Comp. Sci.',54),
-> ('55739','Sanchez','Music',38),
-> ('70557','Snow','Physics',0),
-> ('76543','Brown','Comp. Sci.',58),
-> ('76653','Aoi','Elec. Eng.',60),
-> ('98765','Bourikas','Elec. Eng.',98),
-> ('98988','Tanaka','Biology',120);
Query OK, 13 rows affected (0.01 sec)
Records: 13 Duplicates: 0 Warnings: 0
```

5. Insert data into Course table with the given data.

<i>course_id</i>	<i>title</i>	<i>dept_name</i>	<i>credits</i>
BIO-101	Intro. to Biology	Biology	4
BIO-301	Genetics	Biology	4
BIO-399	Computational Biology	Biology	3
CS-101	Intro. to Computer Science	Comp. Sci.	4
CS-190	Game Design	Comp. Sci.	4
CS-315	Robotics	Comp. Sci.	3
CS-319	Image Processing	Comp. Sci.	3
CS-347	Database System Concepts	Comp. Sci.	3
EE-181	Intro. to Digital Systems	Elec. Eng.	3
FIN-201	Investment Banking	Finance	3
HIS-351	World History	History	3
MU-199	Music Video Production	Music	3
PHY-101	Physical Principles	Physics	4

```
mysql> INSERT INTO COURSE VALUES
-> ('BIO-101','Intro. to Biology','Biology',4),
-> ('BIO-301','Genetics','Biology',4),
-> ('BIO-399','Computational Biology','Biology',3),
-> ('CS-101','Intro. to Computer Science','Comp. Sci.',4),
-> ('CS-190','Game Design','Comp. Sci.',4),
-> ('CS-315','Robotics','Comp. Sci.',3),
-> ('CS-319','Image Processing','Comp. Sci.',3),
-> ('CS-347','Database System Concepts','Comp. Sci.',3),
-> ('EE-181','Intro. to Digital Systems','Elec. Eng.',3),
-> ('FIN-201','Investment Banking','Finance',3),
-> ('HIS-351','World History','History',3),
-> ('MU-199','Music Video Production','Music',3),
-> ('PHY-101','Physical Principles','Physics',4);
Query OK, 13 rows affected (0.01 sec)
Records: 13 Duplicates: 0 Warnings: 0
```

6. Insert data into Section table with the given data.

<i>course_id</i>	<i>sec_id</i>	<i>semester</i>	<i>year</i>	<i>building</i>	<i>room_number</i>
BIO-101	1	Summer	2009	Painter	514
BIO-301	1	Summer	2010	Painter	514
CS-101	1	Fall	2009	Packard	101
CS-101	1	Spring	2010	Packard	101
CS-190	1	Spring	2009	Taylor	3128
CS-190	2	Spring	2009	Taylor	3128
CS-315	1	Spring	2010	Watson	120
CS-319	1	Spring	2010	Watson	100
CS-319	2	Spring	2010	Taylor	3128
CS-347	1	Fall	2009	Taylor	3128
EE-181	1	Spring	2009	Taylor	3128
FIN-201	1	Spring	2010	Packard	101
HIS-351	1	Spring	2010	Painter	514
MU-199	1	Spring	2010	Packard	101
PHY-101	1	Fall	2009	Watson	100

```
mysql> INSERT INTO SECTION VALUES
-> ('BIO-101',1,'Summer',2009,514,'Painter'),
-> ('BIO-301',1,'Summer',2010,514,'Painter'),
-> ('CS-101',1,'Fall',2009,101,'Packard'),
-> ('CS-101',1,'Spring',2010,101,'Packard'),
-> ('CS-190',1,'Spring',2009,3128,'Taylor'),
-> ('CS-190',2,'Spring',2009,3128,'Taylor'),
-> ('CS-315',1,'Spring',2010,120,'Watson'),
-> ('CS-319',1,'Spring',2010,100,'Watson'),
-> ('CS-319',2,'Spring',2010,3128,'Taylor'),
-> ('CS-347',1,'Fall',2009,3128,'Taylor'),
-> ('EE-181',1,'Spring',2009,3128,'Taylor'),
-> ('FIN-201',1,'Spring',2010,101,'Packard'),
-> ('HIS-351',1,'Spring',2010,514,'Painter'),
-> ('MU-199',1,'Spring',2010,101,'Packard'),
-> ('PHY-101',1,'Fall',2009,100,'Watson');
```

Query OK, 15 rows affected (0.01 sec)
Records: 15 Duplicates: 0 Warnings: 0

7. Insert data into Classroom table with the given data.

<i>building</i>	<i>room_number</i>	<i>capacity</i>
Packard	101	500
Painter	514	10
Taylor	3128	70
Watson	100	30
Watson	120	50

```
mysql> INSERT INTO CLASSROOM VALUES
-> (101,'Packard',500),
-> (514,'Painter',10),
-> (3128,'Taylor',70),
-> (100,'Watson',30),
-> (120,'Watson',50);
```

Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

8. Insert data into Teaches table with the given data.

<i>ID</i>	<i>course_id</i>	<i>sec_id</i>	<i>semester</i>	<i>year</i>
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

```
mysql> INSERT INTO TEACHES VALUES
-> ('10101','CS-101',1,'Fall',2009),
-> ('10101','CS-315',1,'Spring',2010),
-> ('10101','CS-347',1,'Fall',2009),
-> ('12121','FIN-201',1,'Spring',2010),
-> ('15151','MU-199',1,'Spring',2010),
-> ('22222','PHY-101',1,'Fall',2009),
-> ('32343','HIS-351',1,'Spring',2010),
-> ('45565','CS-101',1,'Spring',2010),
-> ('45565','CS-319',1,'Spring',2010),
-> ('76766','BIO-101',1,'Summer',2009),
-> ('76766','BIO-301',1,'Summer',2010),
-> ('83821','CS-190',1,'Spring',2009),
-> ('83821','CS-190',2,'Spring',2009),
-> ('83821','CS-319',2,'Spring',2010),
-> ('98345','EE-181',1,'Spring',2009);
Query OK, 15 rows affected (0.01 sec)
Records: 15 Duplicates: 0 Warnings: 0
```

9. Insert data into Takes table with the given data.

ID	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+

```
mysql> INSERT INTO TAKES VALUES
-> ('00128','CS-101',1,'Fall',2009,'A'),
-> ('00128','CS-347',1,'Fall',2009,'A-'),
-> ('12345','CS-101',1,'Fall',2009,'C'),
-> ('12345','CS-190',2,'Spring',2009,'A'),
-> ('12345','CS-315',1,'Spring',2010,'A'),
-> ('12345','CS-347',1,'Fall',2009,'A'),
-> ('19991','HIS-351',1,'Spring',2010,'B'),
-> ('23121','FIN-201',1,'Spring',2010,'C+'),
-> ('44553','PHY-101',1,'Fall',2009,'B-'),
-> ('45678','CS-101',1,'Fall',2009,'F'),
-> ('45678','CS-101',1,'Spring',2010,'B+'),
-> ('45678','CS-319',1,'Spring',2010,'B'),
-> ('54321','CS-101',1,'Fall',2009,'A-'),
-> ('54321','CS-190',2,'Spring',2009,'B+');
Query OK, 14 rows affected (0.01 sec)
Records: 14  Duplicates: 0  Warnings: 0
```

10. Insert data into Advisor table with the given data.

s_id	i_id
00128	45565
12345	10101
23121	76543
44553	22222
45678	22222
76543	45565
76653	98345
98765	98345
98988	76766

```
mysql> INSERT INTO ADVISOR VALUES
      -> ('10101','00128'),
      -> ('12121','23121'),
      -> ('15151','55739'),
      -> ('22222','44553'),
      -> ('32343','19991'),
      -> ('45565','12345'),
      -> ('76766','98988');
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0
```